concept, not a fundamental one. It's a useful approximation that provides genuine physical insight. The separate branches of the wave function aren't put in as part of the basic architecture of the theory. It's just extraordinarily convenient for us human beings to think of a superposition of many such worlds, rather than treating the quantum state as an undifferentiated abstraction."

Her father's eyes widened a bit. "This is worse than I feared. It sounds like you're going to tell me that a 'world' isn't even a well-defined concept in Many-Worlds."

"They're just as well defined as entropy is. If we were a nineteenth-century Laplace demon, who knew the position and momentum of every particle in the universe, we would never have to stoop to defining a coarse-grained notion like 'entropy.' Likewise, if we knew the exact wave function of the universe, we would never have to talk about 'branches.' But in both cases we are poor finite creatures with dramatically incomplete information, and invoking these higher-level concepts is extremely useful."

Alice could tell that her father was losing patience. "I just want to know how many worlds there are," he said. "If you can't answer that, you're not doing a very good sales job here."

"Must be that devotion to honesty under any circumstances that you inculcated into me at a young age," Alice said with a shrug. "It depends on how we divide the quantum state into worlds."

"And isn't there some obvious right way?"

"Sometimes! In simple situations where measurements have a manifestly discrete outcome, like measuring the spin of an electron, we can safely say that the wave function branches in two, and the number of worlds (whatever that was) doubles. When we're measuring a quantity that is in principle continuous, like the position of a particle, things are less well defined. In that case we can define a total weight attached to a certain range of outcomes, the wave function squared, but not an absolute number of branches. That number would depend on how finely we want to subdivide our description of the measurement outcome, which is ultimately a choice that's up to us. One of my favorite quotes along these lines is from David Wallace: 'Asking how many worlds there are is like asking how many experiences you had yesterday, or how many regrets a repentant criminal has had. It makes perfect sense to say that you had many experiences or that